ML Sequence Estimation using Viterbi

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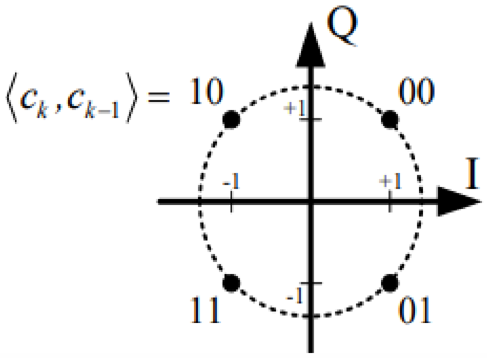
Consider a QPSK system, which discrete-time equivalent channel is given by $p_0 = 1$, $p_1 = 0.3$, and $p_2 = 0.7$, and which equivalent discrete-time noise is a discrete-time WSS Gaussian random process with

$$\mathrm{E}[w_k w_j^*] = \begin{cases} 1, & k = j \\ 0, & else \end{cases}$$

Assuming the constellation is as shown below.



Bit	Symbol
00	1 + j
01	1 - j
10	-1 + j
11	-1 - j

$$r_k = x_k + 0.3\, x_{k-1} + 0.7\, x_{k-2} + w_k$$

AWGN

Assuming that a sequence of QPSK symbols $x_{-2}, x_{-1}, \boxed{x_0, \dots, x_9,} x_{10}, x_{11}$, is transmitted, where x_{-2}, x_{-1}, x_{10} and x_{11} are all known to be $(1+j)$. The corresponding sequences of symbol-spaced received signal symbols are given by

$$\begin{bmatrix} y_0 \\ \vdots \\ y_{11} \end{bmatrix} = \begin{bmatrix} 0.0222 - j0.0169 \\ 1.4509 - j0.7471 \\ -1.6027 - j1.9480 \\ 1.6105 - j2.2362 \\ -1.1898 + j0.1263 \\ -0.6067 + j0.8870 \\ 0.0579 - j0.1223 \\ 0.6309 - j0.4483 \\ -0.0330 + j0.2217 \\ 1.5283 - j1.6817 \\ 0.4960 + j1.1453 \\ 2.3859 + j0.7010 \end{bmatrix}$$

Cost function

使用 Viterbi 演算法找出最可能的傳送符號序列

題目給的 cost function 是 maximum likelihood，所以 path metric 是

$$\min(J) = \min\left(\sum_{k=0}^{11} |y_k - \hat{y}_k|^2\right)$$

而 branch metric 是

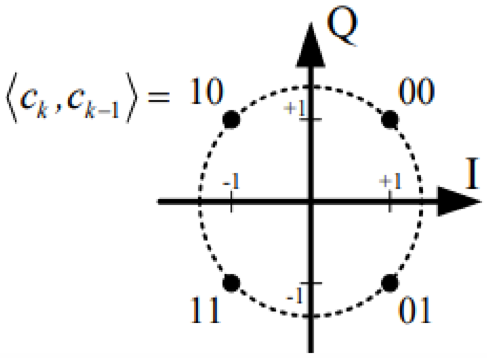
$$j_k = |y_k - \hat{y}_k|^2$$

把一路經過的 branch metric 加起來，就是 path metric

Consider a QPSK system, which discrete-time equivalent channel is given by $p_0 = 1$, $p_1 = 0.3$, and $p_2 = 0.7$, and which equivalent discrete-time noise is a discrete-time WSS Gaussian random process with

$$\mathrm{E}\left[w_k w_j^*\right] = \begin{cases} 1, & k = j \\ 0, & else \end{cases}$$

Assuming the constellation is as shown below.



Bit	Symbol
00	1 + j
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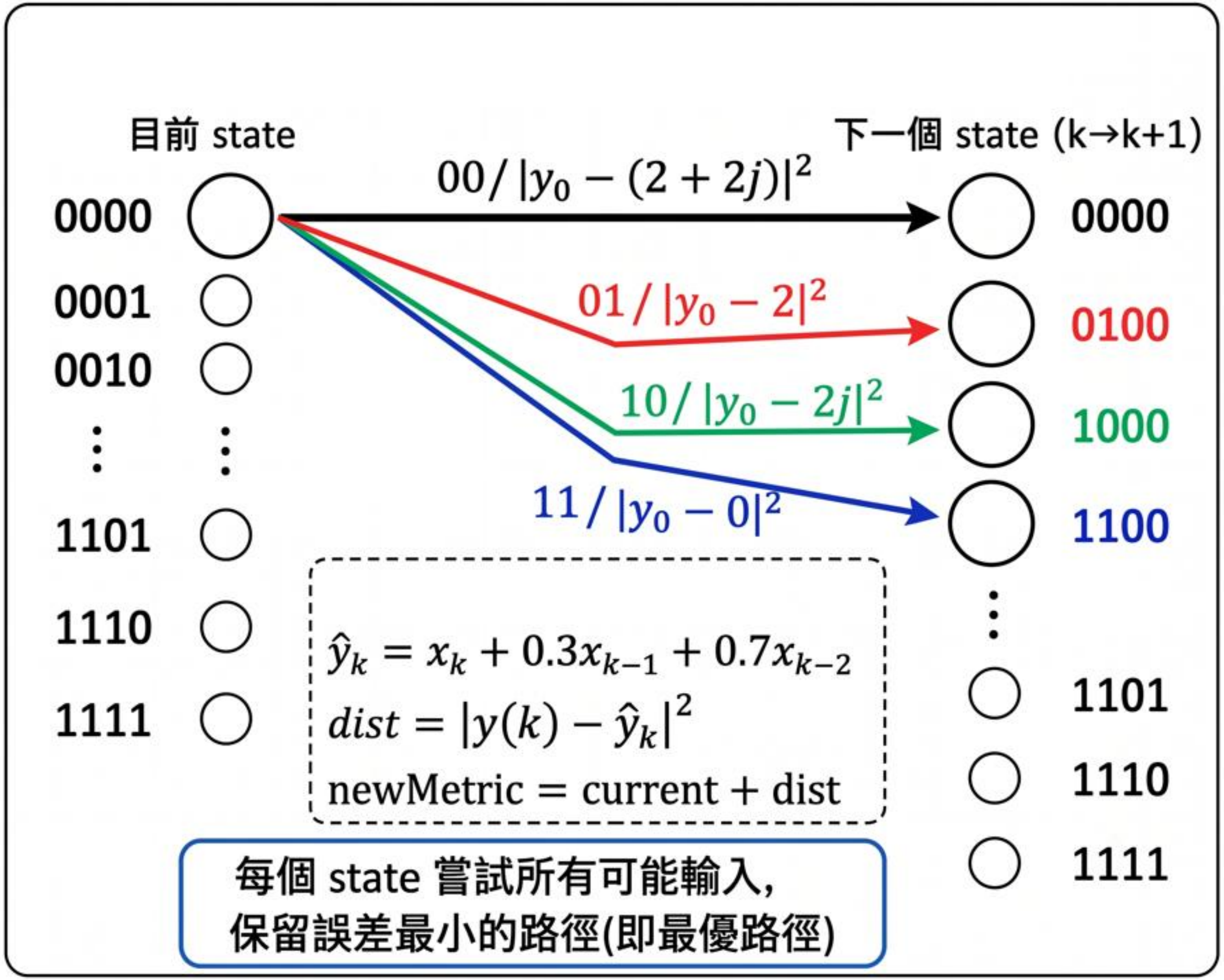
10	-1 + j
11	-1 - j

$$r_k = x_k + 0.3\, x_{k-1} + 0.7\, x_{k-2} + w_k$$

$S_k = (x_{k-1}, x_{k-2}) \rightarrow$ 16state


目前 state
下一個 state (k→k+1)
0000
0001
0010
1101
1110
1111
$00 / |y_0 - (2+2j)|^2$
$01 / |y_0 - 2|^2$
$10 / |y_0 - 2j|^2$
$11 / |y_0 - 0|^2$
0000
0100
1000
1100
1101
1110
1111
$\hat{y}_k = x_k + 0.3x_{k-1} + 0.7x_{k-2}$
$dist = |y(k) - \hat{y}_k|^2$
newMetric = current + dist
每個 state 嘗試所有可能輸入，
保留誤差最小的路徑(即最優路徑)

Algorithm

① 初始化

只考慮初始狀態 $x_{-1} = x_{-2} = 00 \;\Rightarrow\; (a,b) = (0,0) \;\Rightarrow\; \mathrm{id}x = 1$ 的累積誤差為 0。
其他皆為 ∞。

② 主迴圈（time = 1 ~ 12）

- 若 $\mathrm{time} \le 10$：input 可為 0~3 (未知)
- 若 $\mathrm{time} > 10$：已知 $x_{10} = x_{11} = 00 \Rightarrow \mathrm{input} = 0$

對每個目前狀態 (a,b)：

- 嘗試所有輸入
- 計算輸出 $\hat{y}_k$ 、誤差 d、新的累積誤差 newMetric
- 若 newMetric 比 pathMetric(time+1, newIdx) 小，
  就更新 pathMetric, fromState, usedSym

③ 回溯（time = 12 → 1）

從最終狀態 $idx = 1$（因為最後兩個符號已知為 00，所以狀態應回到 (0,0)）
一路往回追溯，取出每一步選擇的輸入符號（0~3）。

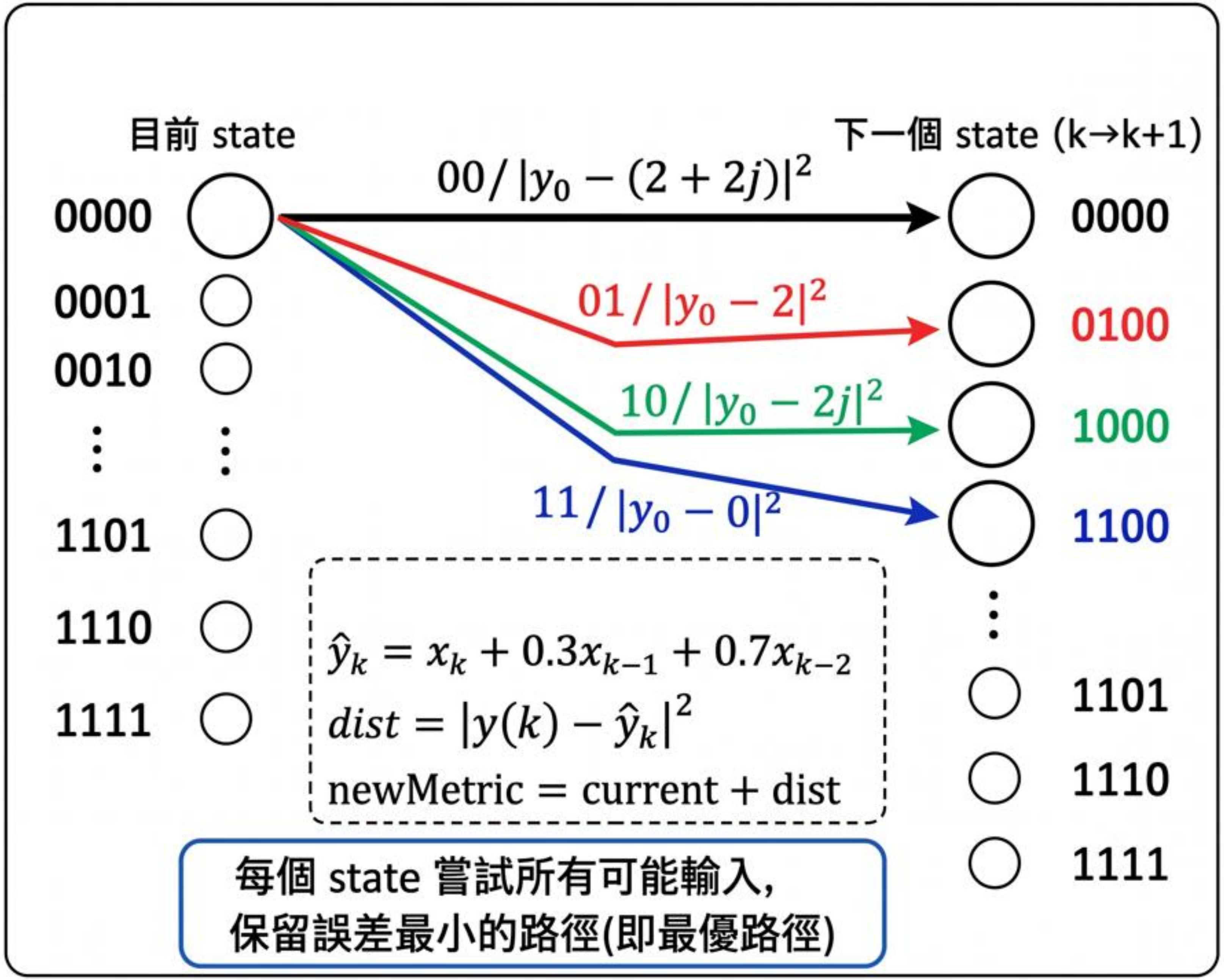
④ 輸出

將 symbolIndex(1~10) 轉成星座點輸出，並印出最小累積誤差。

MATLAB Code

```
symbol = [1+1i; 1-1i; -1+1i; -1-1i];
y = [
     0.0222 - 1i*0.0169
     1.4509 - 1i*0.7471
    -1.6027 - 1i*1.9480
     1.6105 - 1i*2.2362
    -1.1898 + 1i*0.1263
    -0.6067 + 1i*0.8870
     0.0579 - 1i*0.1223
     0.6309 - 1i*0.4483
    -0.0330 + 1i*0.2217
     1.5283 - 1i*1.6817
     0.4960 + 1i*1.1453
     2.3859 + 1i*0.7010
];

state_1 = [0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3];   % xk-1
state_2 = [0 1 2 3 0 1 2 3 0 1 2 3 0 1 2 3];   % xk-2
```


目前 state
下一個 state (k→k+1)
0000
0001
0010
1101
1110
1111
00/ $|y_0-(2+2j)|^2$
01/ $|y_0-2|^2$
10/ $|y_0-2j|^2$
11/ $|y_0-0|^2$
0000
0100
1000
1100
1101
1110
1111
$\hat{y}_k = x_k + 0.3x_{k-1} + 0.7x_{k-2}$
$dist = |y(k) - \hat{y}_k|^2$
newMetric = current + dist
每個 state 嘗試所有可能輸入,
保留誤差最小的路徑(即最優路徑)

MATLAB Code

state (idx)	(a,b) = (x_{k-1}, x_{k-2})	bits(a)	bits(b)	idx = a*4 + b + 1
0	(0, 0)	00	00	0*4 + 0 + 1 = 1
1	(0, 1)	00	01	0*4 + 1 + 1 = 2
2	(0, 2)	00	10	0*4 + 2 + 1 = 3
3	(0, 3)	00	11	0*4 + 3 + 1 = 4
...	...	...	...	...
12	(3, 0)	11	00	3*4 + 0 + 1 = 13
13	(3, 1)	11	01	3*4 + 1 + 1 = 14
14	(3, 2)	11	10	3*4 + 2 + 1 = 15
15	(3, 3)	11	11	3*4 + 3 + 1 = 16

MATLAB Code

```
pathMetric = inf(16, 13);    % pathMetric(idx, time+1)
fromState  = zeros(16, 12);  % fromState(newIdx, time)
usedSym    = zeros(16, 12);  % usedSym(newIdx, time)

% 初始狀態  x(-1) = x(-2) = 0
pathMetric(1, 1) = 0;
```

```
for time = 1:12
    if time <= 10
        tryInputs = 0:3;
    else
        tryInputs = 0;       % x(10) = x(11) = 0
    end

    [redacted]

end
```

```
for time = 1:12
    if time <= 10
        tryInputs = 0:3;
    else
        tryInputs = 0;       % x(10) = x(11) = 0
    end

    for idx = 1:16
        a = state_1(idx);     % x(k-1)
        b = state_2(idx);     % x(k-2)
[redacted]
    end
end
```

```matlab
for time = 1:12
    if time <= 10
        tryInputs = 0:3;
    else
        tryInputs = 0;       % x(10) = x(11) = 0
    end

    for idx = 1:16
        a = state_1(idx);     % x(k-1)
        b = state_2(idx);     % x(k-2)
        current = pathMetric(idx, time);

        for input = tryInputs
            expectedY = symbol(input+1) + 0.3 * symbol(a+1) + 0.7 * symbol(b+1);
            dist = abs(y(time) - expectedY)^2;
            newMetric = current + dist;

            % (a,b) ---> (input,a)
            newIdx = input*4 + a + 1;
            if newMetric < pathMetric(newIdx, time+1)
                pathMetric(newIdx, time+1) = newMetric;
                fromState(newIdx, time) = idx;
                usedSym(newIdx, time) = input;
            end
        end
    end
end
```

Pathmetric

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	0	7.9796	15.8276	15.9576	16.8673	8.5415	7.9810	9.1363	8.8801	8.0231	9.9646	6.3679	0.7346
2	Inf	Inf	8.8235	8.6716	10.0319	3.3882	13.1906	12.0975	3.7469	4.5360	12.4734	0.5755	Inf
3	Inf	Inf	11.6175	17.8780	11.9491	11.5957	4.3382	4.8190	13.2339	10.0671	5.6266	11.1163	Inf
4	Inf	Inf	4.6134	10.5920	5.1137	6.4425	9.5478	7.7802	8.1007	6.5800	8.1353	5.3239	Inf
5	Inf	3.9120	8.8392	9.7656	9.5225	10.6467	13.1290	4.6471	5.0209	10.5099	4.8378	Inf	Inf
6	Inf	Inf	4.2351	4.8796	5.0871	7.8934	17.3682	10.0083	0.3537	9.4228	9.7466	Inf	Inf
7	Inf	Inf	4.6291	11.6860	4.6043	13.7009	9.4862	0.3298	9.3747	12.5539	0.4998	Inf	Inf
8	Inf	Inf	0.0250	6.8000	0.1689	10.9477	13.7255	5.6910	4.7075	11.4668	5.4085	Inf	Inf
9	Inf	4.0684	17.6312	11.1468	19.3093	5.3823	3.8387	10.9679	13.0037	3.8911	12.0778	Inf	Inf
10	Inf	Inf	10.6271	3.8608	12.4739	0.2290	9.0483	13.9291	7.8705	0.4040	14.5866	Inf	Inf
11	Inf	Inf	15.8211	15.4672	16.7911	10.8365	0.3114	8.7757	16.5432	8.3351	10.1398	Inf	Inf
12	Inf	Inf	8.8170	8.1812	9.9557	5.6833	5.5210	11.7369	11.4100	4.8480	12.6485	Inf	Inf
13	Inf	0.0008	10.6428	4.9548	11.9645	7.4875	8.9867	6.4787	9.1445	6.3779	6.9510	Inf	Inf
14	Inf	Inf	6.0387	0.0688	7.5291	4.7342	13.2259	11.8399	4.4773	5.2908	11.8598	Inf	Inf
15	Inf	Inf	8.8327	9.2752	9.4463	12.9417	5.4594	4.2865	12.6840	10.8219	5.0130	Inf	Inf
16	Inf	Inf	4.2286	4.3892	5.0109	10.1885	9.6987	9.6477	8.0168	9.7348	9.9217	Inf	Inf

Pathmetric

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1	0	7.9796	15.8276	15.9576	16.8673	8.5415	7.9810	9.1363	8.8801	8.0231	9.9646	6.3679	0.7346
2	Inf	Inf	8.8235	8.6716	10.0319	3.3882	13.1906	12.0975	3.7469	4.5360	12.4734	0.5755	Inf
3	Inf	Inf	11.6175	17.8780	11.9491	11.5957	4.3382	4.8190	13.2339	10.0671	5.6266	11.1163	Inf
4	Inf	Inf	4.6134	10.5920	5.1137	6.4425	9.5478	7.7802	8.1007	6.5800	8.1353	5.3239	Inf
5	Inf	3.9120	8.8392	9.7656	9.5225	10.6467	13.1290	4.6471	5.0209	10.5099	4.8378	Inf	Inf
6	Inf	Inf	4.2351	4.8796	5.0871	7.8934	17.3682	10.0083	0.3537	9.4228	9.7466	Inf	Inf
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14	Inf	Inf	6.0387	0.0688	7.5291	4.7342	13.2259	11.8399	4.4773	5.2908	11.8598	Inf	Inf
15	Inf	Inf	8.8327	9.2752	9.4463	12.9417	5.4594	4.2865	12.6840	10.8219	5.0130	Inf	Inf
16	Inf	Inf	4.2286	4.3892	5.0109	10.1885	9.6987	9.6477	8.0168	9.7348	9.9217	Inf	Inf

fromState

	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	4	2	4	4	3	4	2	2	3	2
2	0	5	8	6	8	8	7	7	6	6	7	0
3	0	9	12	10	12	10	11	12	10	10	11	0
4	0	13	16	14	16	14	15	15	14	14	15	0
5	1	1	4	2	4	4	3	3	2	2	0	0
6	0	5	8	6	8	7	7	7	6	6	0	0
7	0	9	12	10	12	10	11	11	10	10	0	0
8	0	13	16	14	16	13	15	15	14	14	0	0
9	1	1	4	2	4	2	3	4	2	2	0	0
10	0	5	8	6	8	6	7	7	6	6	0	0
11	0	9	12	10	12	10	9	10	10	10	0	0
12	0	13	16	14	16	14	13	13	14	14	0	0
13	1	1	4	2	4	2	3	3	2	2	0	0
14	0	5	8	6	8	5	7	7	6	6	0	0
15	0	9	12	10	12	10	9	9	10	10	0	0
16	0	13	16	14	16	13	13	13	14	14	0	0

usedSym

Input: 3 1 3 1 2 2 1 1 2 1 (0 0)

	1	2	3	4	5	6	7	8	9	10	11	12
1	0	0	0	0	0	0	0	0	0	0	0	$\boxed{0}$
2	0	0	0	0	0	0	0	0	0	0	$\boxed{0}$	0
3	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0
5	1	1	1	1	1	1	1	1	1	1	0	0
6	0	1	1	1	1	1	1	$\boxed{1}$	1	1	0	0
7	0	1	1	1	1	1	$\boxed{1}$	1	1	$\boxed{1}$	0	0
8	0	$\boxed{1}$	1	$\boxed{1}$	1	1	1	1	1	1	0	0
9	2	2	2	2	2	2	2	2	2	2	0	0
10	0	2	2	2	$\boxed{2}$	2	2	2	$\boxed{2}$	2	0	0
11	0	2	2	2	2	$\boxed{2}$	2	2	2	2	0	0
12	0	2	2	2	2	2	2	2	2	2	0	0
13	$\boxed{3}$	3	3	3	3	3	3	3	3	3	0	0
14	0	3	$\boxed{3}$	3	3	3	3	3	3	3	0	0
15	0	3	3	3	3	3	3	3	3	3	0	0
16	0	3	3	3	3	3	3	3	3	3	0	0

Result

Input: 3 1 3 1 2 2 1 1 2 1 (0 0)

```
估計出的 x0 ~ x9:
x_0  =  -1 -1j,  bits = 11
x_1  =   1 +1j,  bits = 01
x_2  =  -1 -1j,  bits = 11
x_3  =  +1 -1j,  bits = 01
x_4  =  -1 +1j,  bits = 10
x_5  =  -1 +1j,  bits = 10
x_6  =  +1 -1j,  bits = 01
x_7  =  +1 -1j,  bits = 01
x_8  =  -1 +1j,  bits = 10
x_9  =  +1 -1j,  bits = 01

最小誤差 = 0.73459024
```

Result

Command Window

```
估計出的 x0 ~ x9:
x_0 = -1 -1j, bits = 11
x_1 = +1 -1j, bits = 01
x_2 = -1 -1j, bits = 11
x_3 = +1 -1j, bits = 01
x_4 = -1 +1j, bits = 10
x_5 = -1 +1j, bits = 10
x_6 = +1 -1j, bits = 01
x_7 = +1 -1j, bits = 01
x_8 = -1 +1j, bits = 10
x_9 = +1 -1j, bits = 01

最小誤差 = 0.73459024
brute-force 最小誤差 = 0.73459024
brute-force 序列 = [3 1 3 1 2 2 1 1 2 1]
>>
```

```
for x0 = 0:3
for x1 = 0:3
for x2 = 0:3
for x3 = 0:3
for x4 = 0:3
for x5 = 0:3
for x6 = 0:3
for x7 = 0:3
for x8 = 0:3
for x9 = 0:3

    x = [x0 x1 x2 x3 x4 x5 x6 x7 x8 x9 0 0];

    total = 0;
    for k = 1:12
        if k-1 >= 1
            a = x(k-1);
        else
            a = 0;
        end

        if k-2 >= 1
            b = x(k-2);
        else
            b = 0;
        end

        expectedY = symbol(x(k)+1) + 0.3*symbol(a+1) + 0.7*symbol(b+1);
        total = total + abs(y(k) - expectedY)^2;
    end

    if total < best
        best = total;
        best_seq = [x0 x1 x2 x3 x4 x5 x6 x7 x8 x9];
    end

end
end
end
end
end
end
end
end
end
end
```